STIM Rev 1 Channel

28-Oct-2026

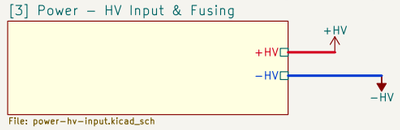
Variant: Preliminary

Rev 2.1.0-dev

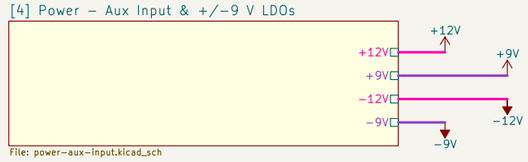
+/-100 V INA149 Howland current-pump stimulator channel: minimal analog path for bench bring-up (UBC ELEC 491, JY063)

POWER: rails leave [3] and [4] as ports; the global power symbols live here. GND is global on every sheet.

[3] Power – HV Input & Fusing



[4] Power – Aux Input & +/−9 V LDOs

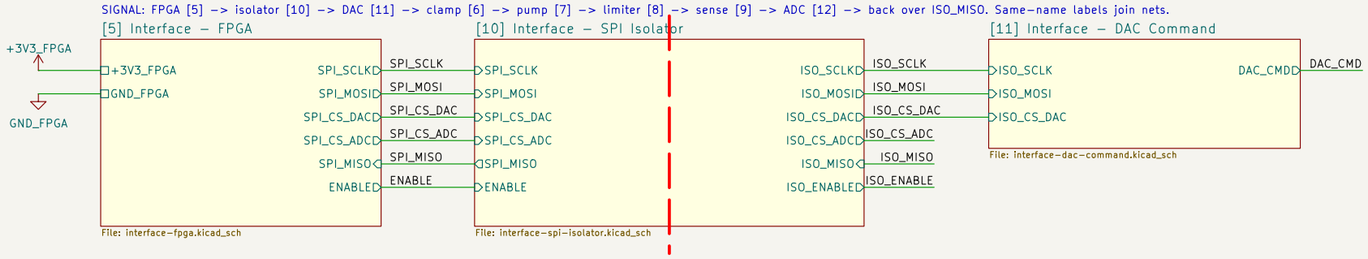


SIGNAL: FPGA [5] -> isolator [10] -> DAC [11] -> clamp [6] -> pump [7] -> limiter [8] -> sense [9] -> ADC [12] -> back over ISO_MISO. Same-name labels join nets.

[5] Interface – FPGA

[10] Interface – SPI Isolator

[11] Interface – DAC Command



ISOLATION BARRIER (inside [10]): left of it J3, +3V3_FPGA, GND_FPGA (FPGA side); right of it the board, GND. Spacing rule set with the isolator part.

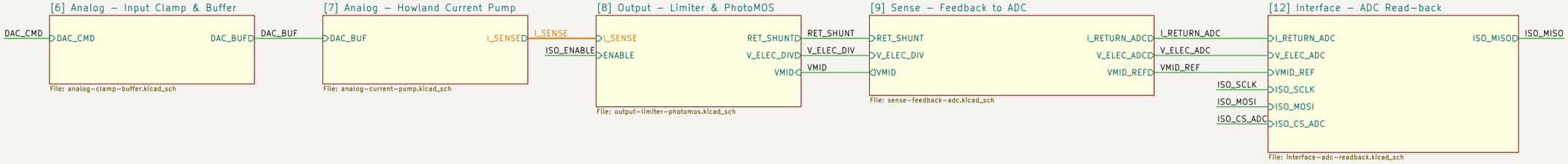
[6] Analog – Input Clamp & Buffer

[7] Analog – Howland Current Pump

[8] Output – Limiter & PhotoMOS

[9] Sense – Feedback to ADC

[12] Interface – ADC Read-back



DESIGN NOTE:
How and why a circuit is sized (grey).

LAYOUT NOTE:
Placement and routing rules from the PCB design reference (blue).

MISC NOTE:
Bench, test and procurement notes (orange). A tag such as D1, L2 or M1 next to a circuit points to the note with that number.

[2] Doc – Revision History



H1 M3 3.4mm NPTH

H2 M3 3.4mm NPTH

H3 M3 3.4mm NPTH

H4 M3 3.4mm NPTH

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Doc - Cover	File Name: STIM_Rev1.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /		Reviewer:	Size: **A3**	Sheet: **1** of **12**

[2] Doc - Revision History

23-OCT-2026 - Initial Capture
Version: 1.0.0 Draft

- Captured the full Rev 1 channel from the simulated netlist and the P4 BoM: 14 sheets, 156 parts.

- Included the full safety chain: over-current and mismatch trips, fault latch, power-on reset, rail monitors, parking switch.

- Netlist matched expected_netlist.csv with 0 differences; ERC clean.

25-OCT-2026 - Minimal Rev 1
Version: 2.0.0-dev Preliminary

- Cut the board to the analog path for bench bring-up: HV input, clamp, buffer, Howland pump, limiter, PhotoMOS, TVS, read-back.

- Removed the trips, latch, power-on reset, rail monitors, enable logic and parking switch (kept in simulation for Rev 2).

- Changed rail shunts RSP/RSN to 10 Ohm for current measurement.

- Changed FPGA connector J3 to DAC_CMD, ENABLE, I_RETURN_ADC, V_ELEC_ADC, VMID_REF and grounds; no power.

- Added the read-back stage (OPA4192) for the FPGA ADC.

- Added bench bias jumpers JP1/JP2, current-loop header JP3, 29 test loops and M3 mounting holes.

26-OCT-2026 - Schematic Review
Version: 2.0.0-dev Preliminary

- Reduced the low-voltage input to one +12 V (J2, 2-pin screw terminal); U32 isolated DC-DC makes +/-12 V on board.

- Removed +5 V; LM4040 bias R50 now from +9 V (5.1k).

- Changed U1 and U3 from OPA171 to OPA192: output accuracy passes without the zero calibration.

- Added red LEDs: PWR (D8), HV LIVE (D7) and OUT EN (D6, in the PhotoMOS LED string; RLED1 now 750 Ohm).

- Fixed the AQY214S symbol (its pins were hidden).

- Resized passives: 0603 by default; 0805/1206/2512 only where voltage, power or precision needs it; one part per value.

- Wired the top sheet; ERC clean; page numbers now follow the sheet titles.

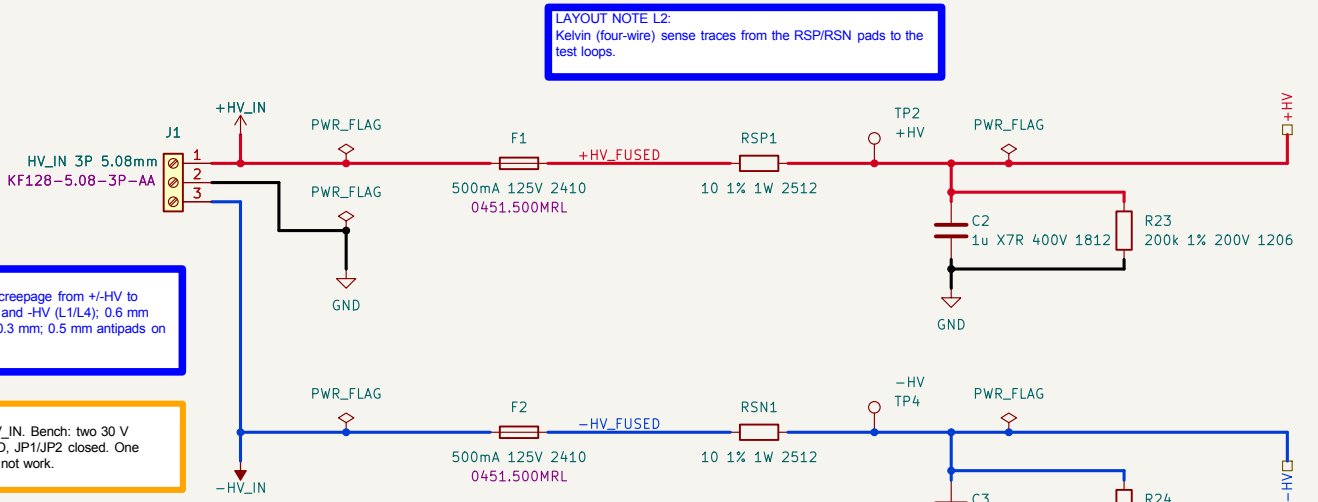
- Formatted after the FoC Motor Controller and AMULET: cover index, block frames with captions, this page. PCB text in JetBrains Mono; 3D model for every part.

DD-MMM-YYYY - Next Revision
Version: _____

- _____
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Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Doc - Revision History	File Name: revision-history.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[2] Doc - Revision History/		Reviewer:	Size: A4	Sheet: 2 of 12

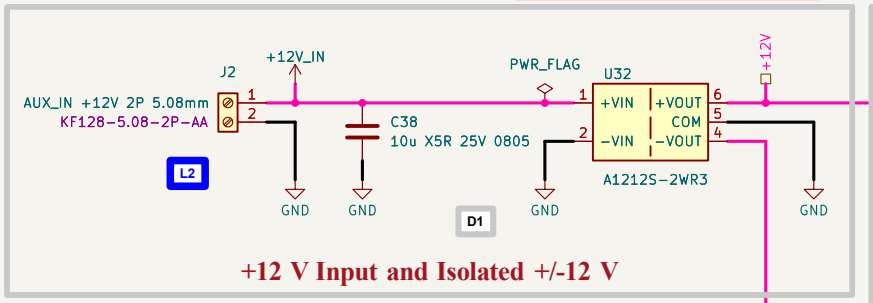
[3] Power - HV Input & Fusing



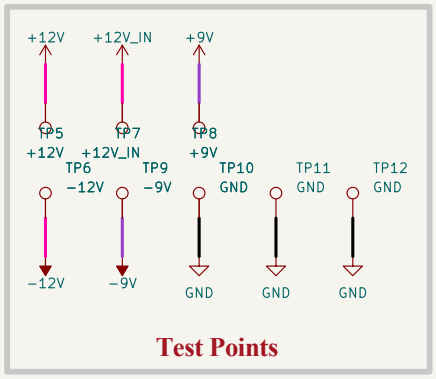
Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Power - HV Input & Fusing	File Name: power-hv-input.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[3] Power - HV Input & Fusing/		Reviewer:	Size: **A4**	Sheet: **3** of **12**

[4] Power - Aux Input & +/-9 V LDOs

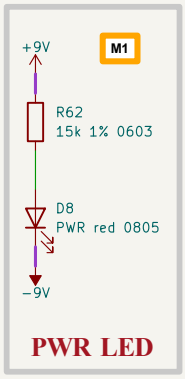
CONFIRM GND AND ISO_GND



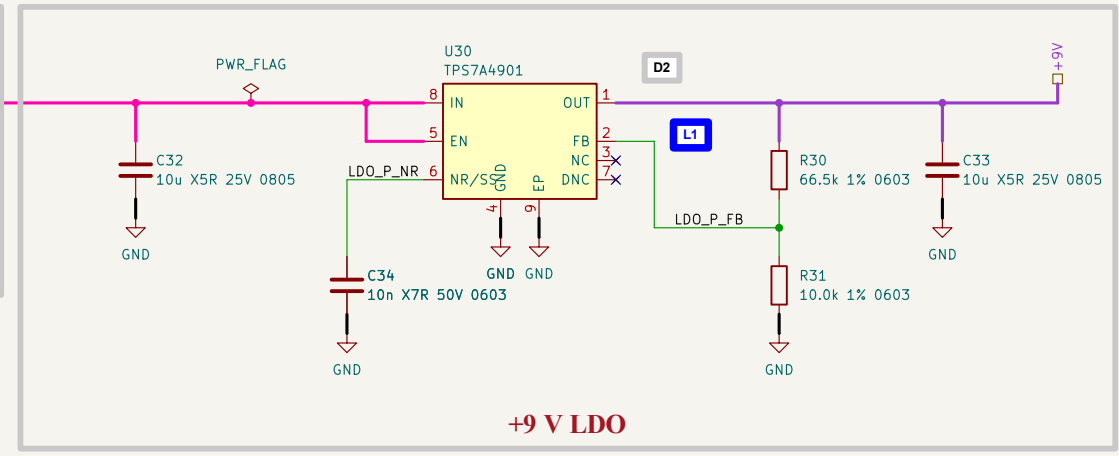
+12 V Input and Isolated +/-12 V



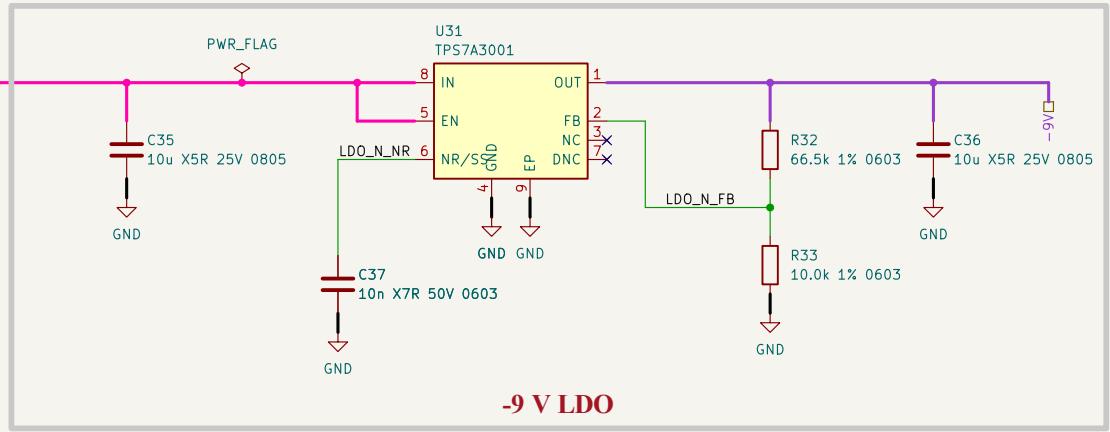
Test Points



PWR LED



+9 V LDO



-9 V LDO

DESIGN NOTE D1:
U32 A1212S-2WR3 (YLPTEC): input 10.8-13.2 V. Pinout per the YLPTEC datasheet: check the module label. The -12 V load (~7 mA) is under the 10 % minimum load, so that rail may sit up to 10 % high: fine for U31 and U22.

DESIGN NOTE D2:
+/-9 V sets the buffer ceiling (D-01): 1.188 V x (1 + 66.5k / 10k) = 9.09 V (9.00-9.27 V). Changing R30-R33 changes the zener-open current ceiling (9.15 mA): re-check D-01 first.

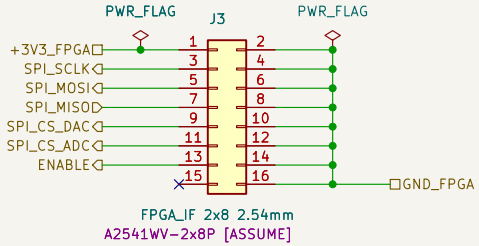
LAYOUT NOTE L1:
Thermal pad (pin 9) to GND with vias; CIN, COUT and the NR/SS cap beside their pins. 0.65 mm pitch: 0.15 mm pad gaps (rule "fine-pitch LDO pads"), mask bridges allowed.

LAYOUT NOTE L2:
J2 and U32 on the top edge (LV side); C38 at the U32 input pins.

MISC NOTE M1:
PWR LED (D8) lights only when +12V_IN, U32 and both LDOs are up: the first check at power-up.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Power - Aux Input & +/-9 V LDOs	File Name: power-aux-input.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[4] Power - Aux Input & +/-9 V LDOs/		Reviewer:	Size: A4	Sheet: 4 of 12

[5] Interface - FPGA



DESIGN NOTE D1:
J3 2x8 IDC: 1 +3V3_FPGA, 3 SPI_SCLK, 5 SPI_MOSI, 7 SPI_MISO, 9 SPI_CS_DAC, 11 SPI_CS_ADC, 13 ENABLE (3.3 V; low or open = switches open), 15 spare GPIO for the safety logic (NC now); even pins GND_FPGA. All nets here are FPGA-side; they cross to the board only through [10]. [ASSUME] +3V3_FPGA from the FPGA powers isolator side 1 (ICD_J3 said no power on J3: decide and update the ICD). Analog DAC_CMD / read-back pins are gone: [11] and [12] do that.

LAYOUT NOTE L1:
J3 on the left (LV) edge, on the FPGA side of the isolation barrier with side 1 of [10]. Every signal has a GND_FPGA neighbour on the ribbon. GND_FPGA is its own pour under J3 and the isolator side 1, not board GND.

MISC NOTE M1:
FPGA: hold the DAC at 0 for 3 ms after ENABLE (D-08; the AQY214S closes in 0.5 ms, so this can shorten); clear the DAC when ENABLE drops; one SPI bus, two chip selects (DAC [11], ADC [12]); subtract VMID_REF from both read-back channels in software.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - FPGA	File Name: interface-fpga.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[5] Interface - FPGA/		Reviewer:	Size: **A4**	Sheet: **5** of **12**

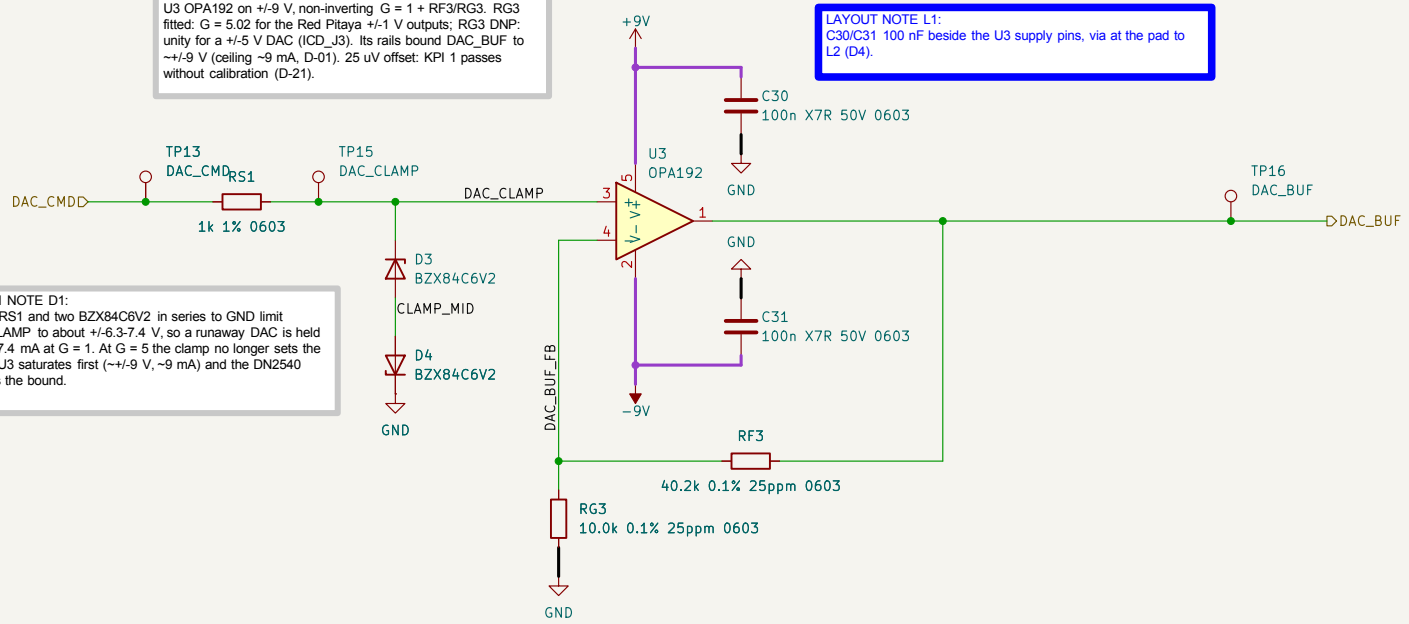
[6] Analog - Input Clamp & Buffer

MISC NOTE M1:
TP13 DAC_CMD -> TP15 DAC_CLAMP -> TP16 DAC_BUF: probe all three to see the clamp act.

DESIGN NOTE D2:
U3 OPA192 on +/-9 V, non-inverting G = 1 + RF3/RG3. RG3 fitted: G = 5.02 for the Red Pitaya +/-1 V outputs; RG3 DNP: unity for a +/-5 V DAC (ICD_J3). Its rails bound DAC_BUF to ~+/-9 V (ceiling ~9 mA, D-01). 25 uV offset: KPI 1 passes without calibration (D-21).

LAYOUT NOTE L1:
C30/C31 100 nF beside the U3 supply pins, via at the pad to L2 (D4).

DESIGN NOTE D1:
Clamp: RS1 and two BZX84C6V2 in series to GND limit DAC_CLAMP to about +/-6.3-7.4 V, so a runaway DAC is held to ~6.3-7.4 mA at G = 1. At G = 5 the clamp no longer sets the ceiling: U3 saturates first (~+/-9 V, ~9 mA) and the DN2540 limiter is the bound.



Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Analog - Input Clamp & Buffer	File Name: analog-clamp-buffer.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[6] Analog - Input Clamp & Buffer/		Reviewer:	Size: **A4**	Sheet: **6** of **12**

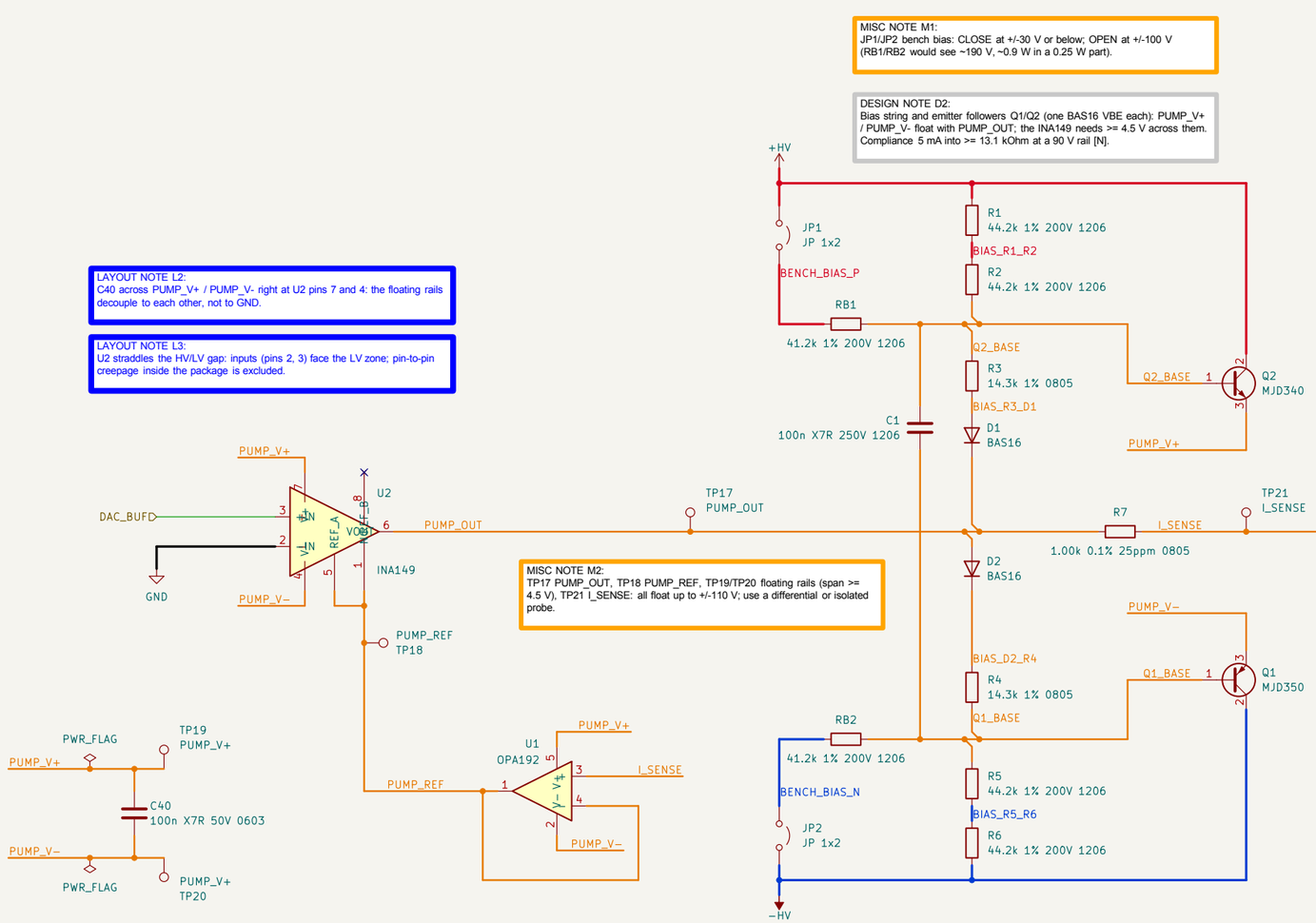
[7] Analog - Howland Current Pump

MISC NOTE M1:
JP1/JP2 bench bias: CLOSE at +/-30 V or below; OPEN at +/-100 V (RB1/RB2 would see ~190 V, ~0.9 W in a 0.25 W part).

DESIGN NOTE D2:
Bias string and emitter followers Q1/Q2 (one BAS16 VBE each): PUMP_V+ / PUMP_V- float with PUMP_OUT; the INA149 needs >= 4.5 V across them. Compliance 5 mA into >= 13.1 kOhm at a 90 V rail [N].

LAYOUT NOTE L2:
C40 across PUMP_V+ / PUMP_V- right at U2 pins 7 and 4: the floating rails decouple to each other, not to GND.

LAYOUT NOTE L3:
U2 straddles the HV/LV gap: inputs (pins 2, 3) face the LV zone; pin-to-pin creepage inside the package is excluded.

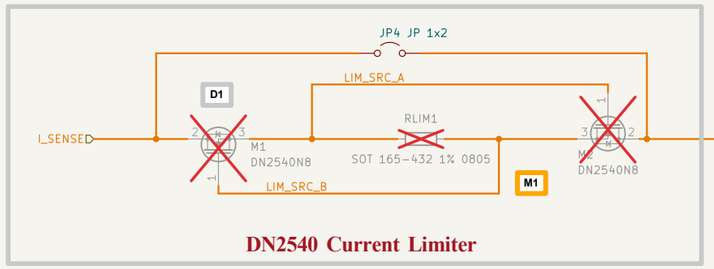


DESIGN NOTE D1:
1 mA/V set by R7 = 1.00 k 0.1 % 25 ppm (0805 for self-heating). U2 INA149 senses across R7; U1 OPA192 buffers I_SENSE back to the REF pins (D-05).

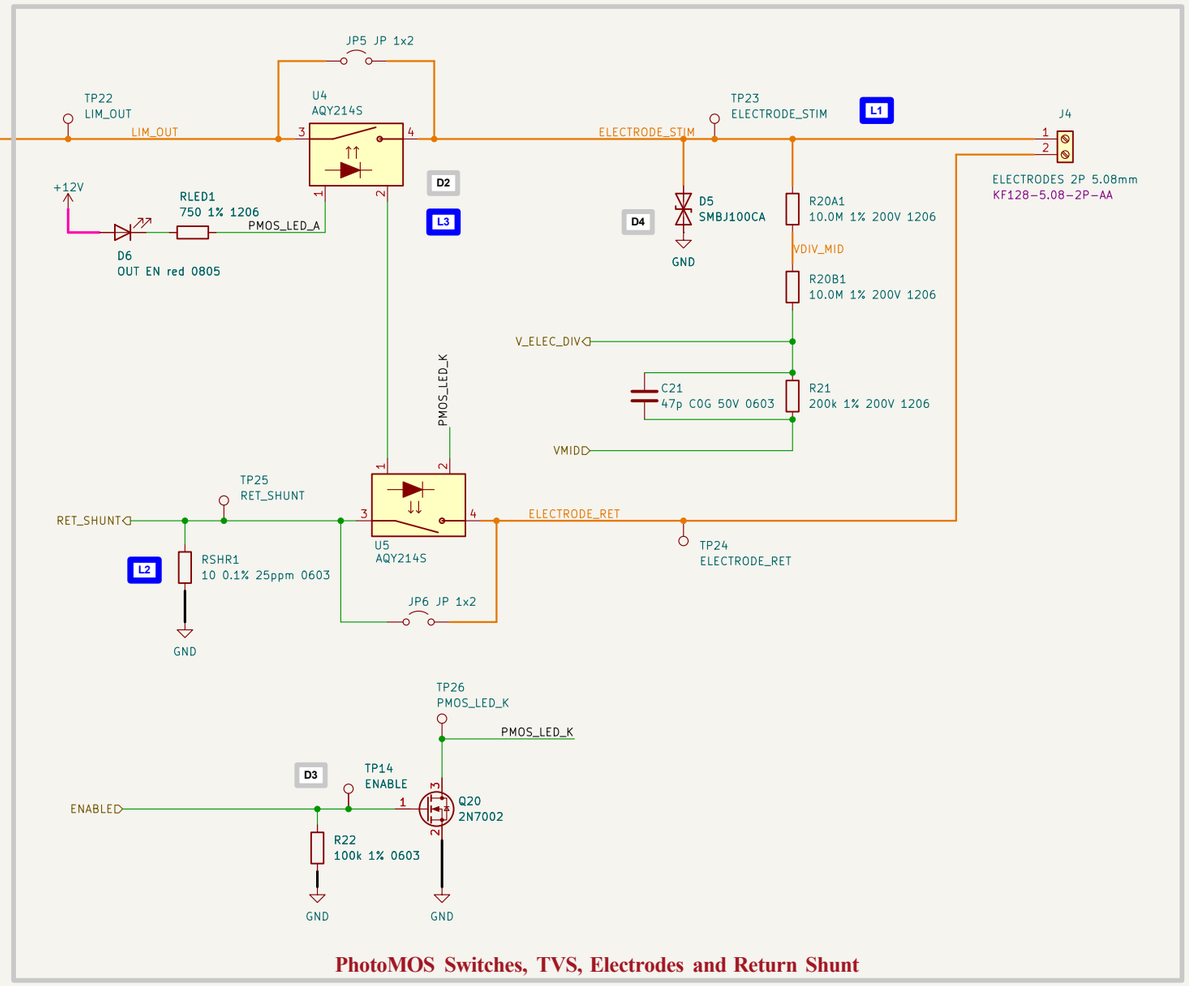
LAYOUT NOTE L1:
Q1/Q2 DPAK: >= 300 mm^2 L1 pour on each collector tab (D6; up to 1.1 W in a 10 mA fault), 1.5 mm to everything; no thermal vias into L2.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Analog - Howland Current Pump	File Name: analog-current-pump.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[7] Analog - Howland Current Pump/		Reviewer:	Size: **A3**	Sheet: **7** of **12**

[8] Output - Limiter & PhotoMOS



DN2540 Current Limiter



PhotoMOS Switches, TVS, Electrodes and Return Shunt

DESIGN NOTE D1:
Limiter (D-02): matched DN2540 pair, one shared RLIM selected on test for 7.6 mA: R = (|Vgs at 7.6 mA| + 10 mV) / 7.6 mA, nearest E96. Limit 6.4-8.8 mA over spread and 0-50 C.

DESIGN NOTE D2:
U4 (stimulating) and U5 (return) share one LED string: one failure opens both. D6 OUT EN is in the string and lights only while the electrodes are live; RLED1 750 Ohm sets ~10 mA.

DESIGN NOTE D3:
ENABLE low or open (R22 pull-down) = LED string off = both switches open.

DESIGN NOTE D4:
D5 SMBJ100CA: VRWM 100 V, VBR 111 V min, above the 110 V rail.

LAYOUT NOTE L1:
Divider: R20A1 in the HV island, R20B1 straddles the gap (V_ELEC_DIV pad on the LV side), R21/C21 on the LV side next to it.

LAYOUT NOTE L2:
RSHR: Kelvin traces from its pads to U22 (0.1 % part, D5); keep the return path J4 -> U5 -> RSHR -> GND short.

LAYOUT NOTE L3:
PhotoMOS straddle the gap: LED pins (1, 2) toward the LV zone, switch pins (3, 4) toward HV.

MISC NOTE M1:
Buy ~10 DN2540N8 per channel; pair them within 100 mV of Vgs at 7.6 mA. RLIM1 is 0805 for hand rework.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Output - Limiter & PhotoMOS	File Name: output-limiter-photomos.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[8] Output - Limiter & PhotoMOS/		Reviewer:	Size: **A3**	Sheet: **8** of **12**

[9] Sense - Feedback to ADC

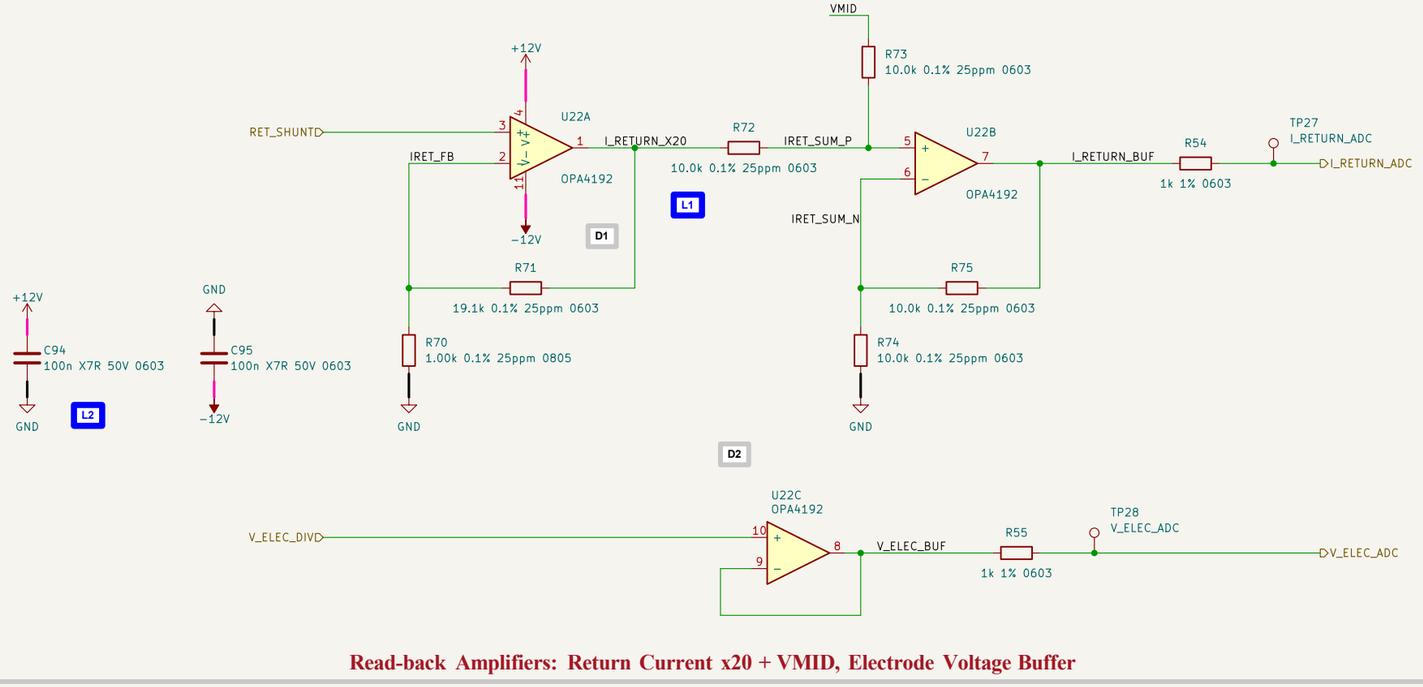
DESIGN NOTE D1:
I_RETURN_ADC = VMID + 0.201 V/mA: RSHR 10 mV/mA x 20.1 (U22A, R71/R70) + VMID (U22B, R72-R75 0.1 %). +/-5 mA -> 0.66-2.66 V.

DESIGN NOTE D2:
V_ELEC_ADC = VMID + V_elec / 101 (20 MOhm / 200 k returned to VMID); U22C buffers it (20 pA bias).

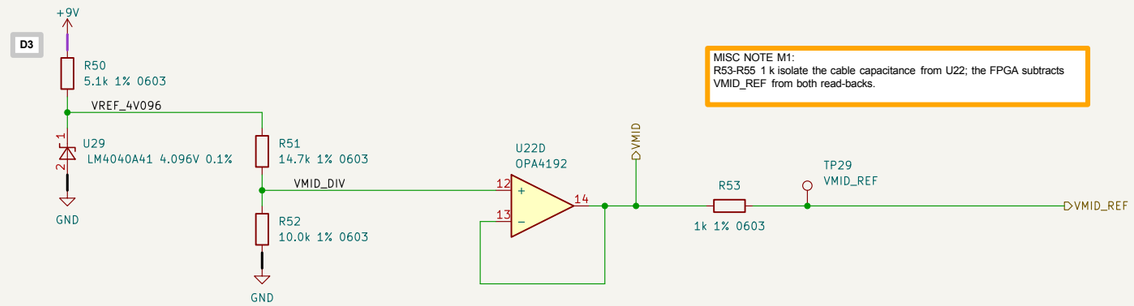
DESIGN NOTE D3:
VMID = 4.096 V x 10.0k / 24.7k = 1.658 V (half of a 3.3 V ADC). Other ADC range: re-select R51/R52. U29 runs at ~0.8 mA from +9 V.

LAYOUT NOTE L1:
0.1 % set R72-R75: short, symmetric traces; no copper pour between their pads.

LAYOUT NOTE L2:
C94/C95 at U22 pins 4 and 11, via at the pad.



Read-back Amplifiers: Return Current x20 + VMID, Electrode Voltage Buffer



VMID Reference

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Sense - Feedback to ADC	File Name: sense-feedback-adc.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[9] Sense - Feedback to ADC/		Reviewer:	Size: **A3**	Sheet: **9** of **12**

[10] Interface - SPI Isolator

FPGA side: J3, +3V3_FPGA / GND_FPGA

SPI_SCLK
SPI_MOSI
SPI_MISO
SPI_CS_DAC
SPI_CS_ADC
ENABLE

Board side: DAC / ADC / ENABLE, board GND

ISO_SCLK
ISO_MOSI
ISO_MISO
ISO_CS_DAC
ISO_CS_ADC
ISO_ENABLE

ISOLATION BARRIER:
No copper crosses except the isolator. Creepage / clearance per IEC 60664-1 for the working voltage (+/-100 V electrode side), set with the part (hv-insulation).

DESIGN NOTE D1:
Ports only: no parts yet. One SPI bus from the FPGA crosses the barrier: SCLK, MOSI and the two chip selects (DAC, ADC) go FPGA -> isolated side, MISO comes back. Left side is FPGA ground (J3), right side is the analog ground of the DAC/ADC. Part selection decides: digital isolator (4 or 6 channels) vs. isolator with built-in isolated supply, and where the isolated rail for the converters comes from (U32 A1212S has its COM tied to GND today). ENABLE crosses here too (ISO_ENABLE drives the PhotoMOS on [8]); spare channels for the safety-logic GPIO come later. Side 1 is powered from +3V3_FPGA / GND_FPGA on J3 [5] (decision pending: ICD_J3 said no power on J3). Six channels: five FPGA -> board, MISO back. Rev 1 fallback: 0 Ohm links in place of the isolator, which makes GND_FPGA = GND.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - SPI Isolator	File Name: interface-spi-isolator.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[10] Interface - SPI Isolator/		Reviewer:	Size: **A4**	Sheet: **10** of **12**

[11] Interface - DAC Command

ISO_SCLK

ISO_MOSI

ISO_CS_DAC

DAC_CMD

DESIGN NOTE D1:
Ports only: no parts yet. Serial DAC on the isolated side; its output is DAC_CMD into U3 on [6] (G = 5.02 with RG3 fitted, so a +/-1 V full scale; RG3 DNP for a +/-5 V DAC). Bipolar output vs. unipolar + offset, reference, LDAC/CLR handling and the isolated supply come with the part. DAC_CMD is driven from this sheet only; J3 [5] carries SPI, not the analog command.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:	
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**		
Sheet Title: Interface - DAC Command	File Name: interface-dac-command.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev	
Sheet Path: /[11] Interface - DAC Command/		Reviewer:	Size: **A4**	Sheet: **11** of **12**	

[12] Interface - ADC Read-back

I_RETURN_ADC

V_ELEC_ADC

VMID_REF

ISO_SCLK

ISO_MOSI

ISO_CS_ADC

ISO_MISO

DESIGN NOTE D1:
Ports only: no parts yet. Serial ADC on the isolated side reading the [9] outputs: I_RETURN_ADC = VMID + 0.2 V/mA and V_ELEC_ADC = VMID + V_elec/101, both sized for a 0-3.3 V input; VMID_REF (1.66 V) is available as the mid-scale reference. Channel count, resolution, sample rate, CONVST/DRDY lines and the isolated supply come with the part. Replaces the Red Pitaya slow inputs: the read-back leaves the board over SPI (ISO_MISO through [10]), J3 [5] carries no analog.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - ADC Read-back	File Name: interface-adc-readback.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[12] Interface - ADC Read-back/		Reviewer:	Size: **A4**	Sheet: **12** of **12**